

Unknown generator point Order - security risks

1. Small subgroup attacks - if the order n is small or has small prime factors, an attacker can confine the ECDH secret to a tiny subgroup recovering private keys with minimal computation (Pohlig - Hellman)

2. Without knowing n , the implementation cannot validate that a public key ~~to~~ lie on the correct curve / subgroup. A malicious ~~can~~ on a weak & twist or wrong subgroup leaks private keys bit

point

3. Failure of cryptographic
hardware guarantees - the
security of ECC relies on
ECDLP being hard, which
requires n to be a large
prime. An ~~unknown~~
unknown order may be
composite or smooth makes
DLP tractable via Pollard-Hellman

4. No cofactor mitigation
Standard curves publish
the co-factor $h = \#E/n$
without knowing n , cofactor
is unknown so cofactor
multiplication cannot be
applied correctly

5. Non-compliance with
security standards such
as NIST